Proceedings of the Seventh International Conference on Communication and Electronics Systems (ICCES 2022)
IEEE Xplore Part Number: CFP22AWO-ART; ISBN: 978-1-6654-9634-6

2022 7th International Conference on Communication and Electronics Systems (ICCES) | 978-1-6654-9634-6/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICCES54183.2022.9836021

Securing Web User Privacy with Steganalysis of Images using Deep Learning

Anuratha K,
Department of Information Technology,
Sri Sairam Institute of Technology,
Chennai, India.
anuratha.it@sairamit.edu.in

Dr. J.M Nandhini,
Department of Information Technology,
Sri Sairam Institute of Technology,
Chennai, India.
nandhini.it@sairamit.edu.in

Devadharshini Madhavan
Department of Information Technology,
Sri Sairam Institute of Technology,
Chennai, India.
i8it026@sairamtap.edu.in

Harini A
Department of Information Technology,
Sri Sairam Institute of Technology,
Chennai, India.
i8it022@sairamtap.edu.in

Pavethra Arulmani
Department of Information Technology,
Sri Sairam Institute of Technology,
Chennai, India.
i8it055@sairamtap.edu.in

***Abstract* — Nowadays, images are extensively used in all aspects of community because of their ease of understandability, like cyber forensics and research-based exploration. Tampering digital images in special fields maliciously, may change the information contained therein, form false information, and result in damage to society. Data security is very important when sensitive data is transmitted over the internet. Whenever a website is opened, there are a lot of advertisements displayed on the page. Although they don't intrude on our browsing experience, some images may hide vicious data inside them by using steganography. Hackers have become proficient in embedding their harmful code inside images. As soon as the user clicks on the image, the hidden code starts to affect the host machine by running in the background without the knowledge of the user. Henceforth, this research work proposes a system by using Deep Learning (DL) to detect steganography (Blind Steganalysis). Convolutional Neural Networks (CNN) is used in our system to prognosticate the presence of steganography by providing an extension in the web browser. By testing our model, it's inferred that the accuracy in finding hidden content in images shows good results. The proposed model has achieved an accuracy of 89 percentage.**

Keywords—Steganography, Steganalysis, Blind Steganalysis, Deep Learning, Convolutional Neural Networks

I. INTRODUCTION

Steganography is the method of concealing confidential data inside an ordinary file, image, video, or message to avert detection. In recent times, information or data concealment has become a trending research issue in information security field. Images are the most frequently used media for data or information hiding. Images are made up of pixels. Every pixel generally consists of 8 bits: 1 byte for a black and white image or 24 bits (3 bytes) for a color image-- one byte each for red, green, and blue. Steganography has advanced in today's world so much that a greater part of individuals is beginning to ponder how to reverse the course of steganography. This approach is termed as steganalysis. Steganalysis focuses on detecting images and modifying messages hidden using steganography. Therefore, steganography and steganalysis are said to be two sides of the same coin. Blind steganalysis focuses on detection of secretly embedded messages without providing any insight on what algorithm has been used to hide them. It is mostly preferred when there is no information regarding the encryption algorithm used.

Nowadays, innovations in technology and fast Internet speeds have made information distribution over the world easy and economic. The proposed user privacy protection system is based on the domain of deep learning to ensure security in images using steganalysis, especially, when browsing the internet. Blind steganalysis procedure can be integrated with deep learning to acquire improved outcomes. Steganalysis requires deep learning technique in the form of characterization to decide if a computerized picture contains some secret content. The image dataset is trained and evaluated with the help of feature extraction. Finally, exposing the presence of concealed data in pictures is provided by an extension in the web browser.

II. OBJECTIVES

a) To develop a genuine blind steganalysis method for colour images, that is capable of detecting the existence of hidden information which has been embedded using various steganographic techniques.

b) To develop a secure steganalysis method, using feature extraction and classification techniques using CNN.

c) To assess the proposed method over the existing system.

III. RELATED WORK

There are various papers that have proposed different methodologies to implement steganography detection in images. Steganalysis can be achieved by utilizing machine learning, deep learning or even by simple detection methods. Weike You et al. [1] has adopted a Siamese CNN based architecture that identifies the relationships between the noises of different image sub-regions and shows the output of classification based on them. The experiments were performed using BOSSbase 1.01 and ALASKA #2 to validate the network. The proposed model demonstrates high rate of transferability amidst various image sizes. This paper also tackles the problem by looking at the feasibility of using a network for steganalysis sans updating its variables.

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:04 UTC from IEEE Xplore. Restrictions apply.

Mehdi Boroumand et al. [2] have described a system which provides improved accuracy for blind image steganalysis in JPEG as well as spatial-domain steganography. Here pooling has been disabled in the detector used to computes noise residuals to prevent the repression of stego-signal. Performance can be improved by furnishing the selection channel as second channel. This system detects abnormalities in picture noise patterns in order to locally modified regions.

In the research work [5], Mohamed A. Elshafey et al. has proposed a new CNN approach, called as IGNCNN (Improved Gaussian Convolutional Neural Network) so as to detect steganography in images with low payloads. The pre-processing model has a fixed number of coefficients high pass filter, to minimize the detection error cost. This improved the accuracy, as compared to conventional CNN based algorithms in blind image steganalysis by a fair amount.

Authors Clement Fuji Tsang et al. [14] has improved the already existing architecture of YeNet to detect steganography in the spatial domain. The system makes changes to the payload according to the image size, by using the square root law for constant detectability in order to steganalyze images of different size with a negligible loss. Increase in the size and resolution of images has resulted in lower detection errors. The outcome of the research is a size independent detector which is successful and also precise.

The ideas from some of the above papers and the findings from their implementation served as motivation for our work and provided a platform to improve them. This research work has proposed a system based on the highly used CNN model by incorporating detection algorithms to identify the presence of hidden, embedded by utilizing state-of-the-art steganography techniques WOW and S-UNIWARD.

IV. BACKGROUND

A. *Deep Learning:*

Multilayered networks learn from huge amounts of data in deep learning, a subset of machine learning. Deep learning works with artificial neural networks, which are supposed to mimic humans' thought processes and learning processes; whereas Machine learning involves fundamental theories. Deep Learning helps a computer to learn how to predict and classify information or data by classifying or sorting inputs through layers. It's possible to think of it as a means to automate predictive analytics. While conventional machine learning algorithms are straightforward, deep learning algorithms are assembled in a stack of increasing difficulty and distraction. Image categorization, translation in languages and natural language processing have all benefited from deep learning. It is capable of solving any pattern matching or identification challenge without the need for human intervention.

Deep Learning is based on how the human brain filters the information with the help of neurons. Its primary objective is to mimic sapient-like moral choice. The impulses to conduct the activities are sent by neurons in the brain. In a neural network, artificial neurons link to accomplish tasks like grouping, classification and regression. The unlabeled data is arranged by the neural network based on the data's commonalities or resemblances. Text, images or sound can be used to represent the information gathered which will be the obtained results. Deep learning systems require a vast quantity of data to provide precise outcomes; therefore data is given in the form of enormous data sets.

B. *Steganography:*

Steganography is the approach of concealing confidential facts inside a usual, non-confidential, categories or messages in sequence to keep away from the detection; the concealed facts is then extracted at its target and then the concealed information is usually processed in some manner for you to raise the difficulty of detecting the confidential message. Whilst there are numerous criminal make uses of for steganography, malware inventors have additionally been observed to accomplish steganography to hidden the communication of malicious cipher. When steganography is done on the image, if the user visits the image document they are unable to see the distinction between the actual image document and the ciphered image document. That is fulfilled by saving the information with the Least Significant Bytes (LSB) in the data recorder, which may be achieved manually or with the help of a steganography device. Steganography software is utilised to execute a heterogeneity of features so that it will conceal the information which includes encrypting the facts that allow you to prepare it to be hidden inside another document, keep on track of which bits of the hidden text file containing hidden facts, encrypting the data to be concealed and extracting hidden information by its supposed recipient. There are many steganographic algorithms, some of them are listed below:-

- Spatial –Universal Wavelet Relative Distortion (S-UNIWARD)
- Wavelet Obtained Weights (WOW)
- Hill
- Highly Undetectable Stego (HUGO)

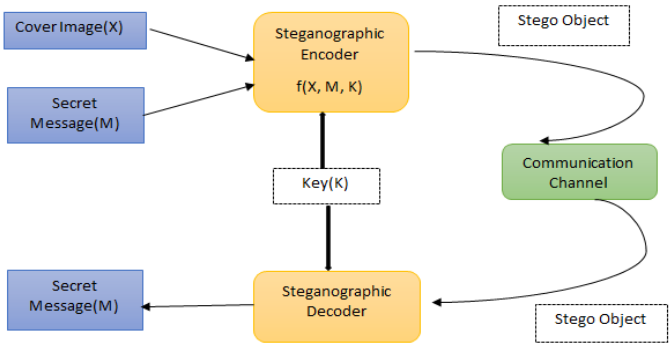

Fig. 1. Steganographic Process

As shown in Fig. 1, the cover image (X) and the secret message (M) are supplied as inputs to the steganographic encoder. The Steganographic Encoder function, f (X, M, K), embeds a secret message inside a cover image. The subsequent Stego - Object is virtually indiscernible from the cover image, with no distinguishable differences. This is the final step in the encryption phase. The Stego - Object is fed into the Steganographic Decoder to recover the cryptic information from the cover image.

C. *Steganalysis:*

The art and science of discovering concealed data in a digital medium is known as Steganalysis. Steganalysis is the process of identifying suspicious streams of information, determining whether they include hidden messages or not, and recovering the secret information. Data and messages can be hidden anywhere on the Internet; that is in the form of text, links, comments, structure, frames, and images. Attacks and checks for undisclosed information can take many formations: classify, extract, disable, or destroy hidden data. The science aids in the construction of the procedure to identify or extract the hidden data, while the art of steganalysis plays a larger part in the selection of aspects of digital data to test for hidden data. The objective of steganalysis is not to support the removal or defusing of substantial covered up information

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:04 UTC from IEEE Xplore. Restrictions apply.

such as copyrights, but to point out approaches that are defenceless and may be misused to examine forbidden secret content.

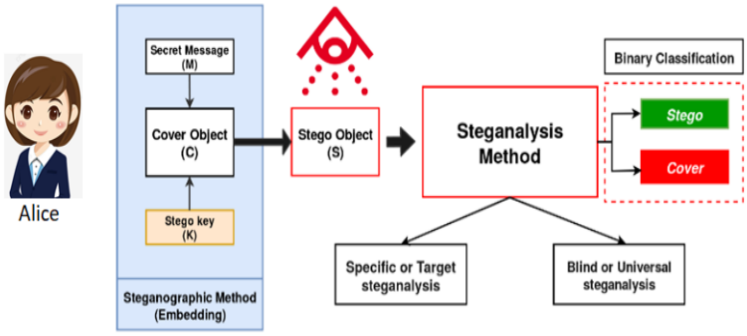

Fig. 2. Steganalysis Model

According to Fig. 2, the secret message is embedded into the cover object using a stego - key (K). The derived target is known as the stego-object (S). This is the final product of steganography on the image to hide the confidential information. This stego - object (S) is fed into the steganalyzer to uncover the hidden content. There are two types of steganalysis, namely blind or universal steganalysis and specific or target steganalysis. Specific steganalysis is used to uncover secret message that is hidden inside the cover object. Blind steganalysis is just utilized to identify in case there's anything covered up within the given stego image or not. It cannot uncover the contents of the concealed information.

D. Blind Steganalysis:

Blind steganalysis, also known as universal steganalysis, may discover concealed information without knowing the content or the embedding method.

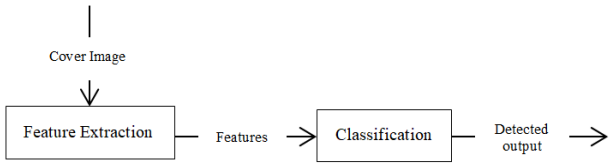

Fig. 3. Steganalysis Process

The steganalysis procedure commences with a cover image that is presumed to contain hidden data, as illustrated in Fig. 3. The specified feature extraction method is used to extract features from the cover image. The obtained features are then fed into a classifier, which ultimately decides if the resulting image holds any embedded data.

E. Convolutional Neural Networks (CNN):

Convolutional neural networks (CNN / ConvNet) are a type of deep neural network that is frequently used to analyze visual images. CNN utilizes dimensional correlations which exist with the training data. Each layer of the neural network connects certain input neurons at the same time, as elucidated in Fig. 4. This region is called a local receptive field which focuses on hidden neurons. CNNs are fully connected feed-forward neural networks that are particularly good at lowering the number of parameters while maintaining model quality. Some strategies are offered in CNN-based steganalysis to circumvent the limitations of traditional steganalysis. These methods can extract information from input images automatically. (Classification and feature extraction are integrated under a sole architecture which makes it feasible to optimize both classification and feature extraction steps simultaneously. CNN-based steganalysis approaches present equivalent results to the conservative approaches.)

The capacity of CNNs to construct an internal representation of a two-dimensional image is one of its advantages. When working with images, this allows the system to operate the location and spectrum structures in the data. In comparison to its predecessors, CNN has the advantage of automatically detecting essential traits without the need for human intervention.

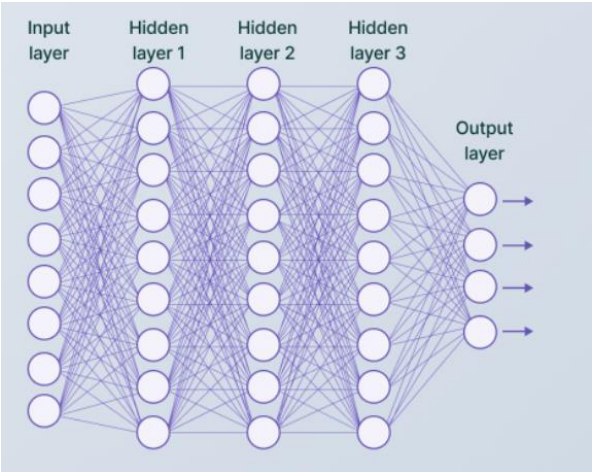

Fig. 4. CNN Model

V. SYSTEM ARCHITECTURE

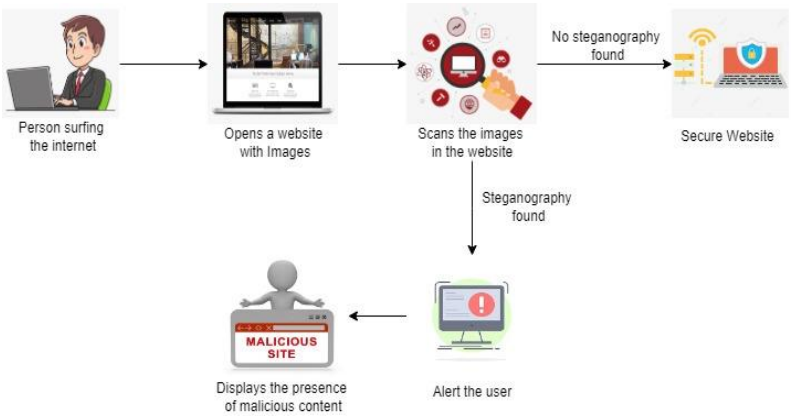

Fig. 5. System Architecture

The model has been designed keeping in mind the security and privacy of Internet users. As represented in Fig. 5, when a person surfs the Internet, he/she is exposed to a lot of images and videos. Most of these videos automatically play in the background, without any intervention from the users. Also, the images present in the websites load on their own. Some of the images may contain unwanted and malicious information embedded in them using various steganographic techniques. As soon as the image loads, it starts executing the hidden code in it, thus compromising the security of the user. At this juncture, our system comes into play.

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:04 UTC from IEEE Xplore. Restrictions apply.

The proposed system scans each and every image present in the website for any hidden data using deep learning algorithms. If it detects even a small piece of information concealed in the image during this process, the system alerts the user by displaying a warning in the webpage. On the other hand, if the system does not find anything during the scanning process, then the system concludes that the page is safe to use by the user. It shows an indication to the user that he/she is secure and can proceed with his browsing (safe website). This scanning process is done every time the user opens a new web page on the Internet.

VI. SYSTEM ANALYSIS

A. *Existing System*

In the previous systems, image steganalysis was implemented using feature extraction and Machine Learning algorithms such as Support Vector Machine (SVM). This approach has been used to expose the existence of steganography. The highest precision achieved by the existing model is 87% for gray scale images and 85% for color images. While there are many algorithms and technologies for steganalysis, but there are no extensions provided in the browsers to detect the presence of steganography. This provides an ease of access for safe browsing to the internet users.

B. *Proposed System*

Steganography is to hide secret data without distortion. Steganalysis is the process of detecting the presence of disguised data. We propose a technique to identify steganography (steganalysis) within images in order to safeguard internet users' privacy from hostile steganographic attacks. By analysing the images displayed in a webpage, we can uncover any embedded data camouflaged in them. If we detect even a small amount of data hidden inside it, we warn the user by displaying an alert "Images displayed here may contain malicious content that could harm your system!". This can be efficiently made possible by using ML to prepare the model with the help of dataset of steganographic images. The model can learn feature extractions by itself with various convolutional layers, in comparison to other existing methods. The feature extraction and classification steps are brought together under a solitary architecture, which implies that the direction of classification can be used during the feature extraction step. The proposed system can be implemented as a webpage extension so that as soon as the user enters a new webpage, the extension will scan for any steganographic content. Our proposed model has an accuracy of 89% for colour images.

VII. SYSTEM IMPLEMENTATION

A. *Modules*

The system implementation consists of 4 modules as described in Fig. 6.

Module 1: Image Dataset Collection
To train the model for blind steganalysis, we have collected a dataset consisting of 10000 images in order to improve accuracy.

Module 2: Feature Selection and Classification
This step has been implemented using Python and CNN. The model is trained using the collected dataset of 10000 images.

Module 3: Testing the System
After classifying the image features, the system detects the presence of any hidden data in the image. Testing the model is important so as to decrease the errors while predicting steganography.

Module 4: Creation of Steganalysis Extension
After successful testing of the system, the next step is to implement the model as an extension in Google Chrome. This becomes an easy way to fortify the privacy of internet users.

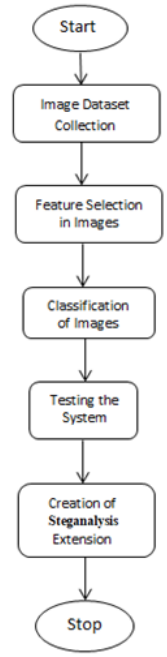

Fig. 6. System Implementation

B. *Dataset Collection*

The dataset consists of 10,000 images collected from BOSSbase 1.01, which are of size 512 × 512 × 1. The images were resized to 256 × 256 × 1 using Matlab. The BOSSbase 1.01 dataset was separated into 4000 pairs training, 1000 pairs for validation and 5000 pairs for testing.

C. *Formulae*

In a neural network, an activation function specifies how the weighted sum of the input is turned into an output from a node or nodes in a layer.

$$3TanH(x) = 3\frac{e^{x}-e^{-x}}{e^{x}+e^{-x}} \quad (1)$$

In the vicinity of the origin, TanH is quite similar to the Y=X function. When the activation function value is low, the matrix operation may be done immediately, making the training process easier. The neural network becomes heavier when both the TanH and sigmoid activation functions are triggered. Therefore, if strong gradients and big learning steps are required, the TanH activation function is used.

VIII. RESULTS AND DISCUSSIONS

Standard machine learning algorithms like Support Vector Machines were initially utilised to overcome the problem of steganography in images. Many Deep Learning (DL) applications, on the other hand, have proven to be the superior option for image processing. Convolutional Neural Networks (CNN) is particularly useful for extracting spatial and frequency features from images.

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:04 UTC from IEEE Xplore. Restrictions apply.

On the BOSSbase 1.01 dataset, the suggested CNN improves accuracy by 8.7% on WOW with 0.2 bpp and 4.4% on WOW with 0.4 bpp, 15.3% on S-UNIWARD with 0.4bpp and 3% on S-UNIWARD with 0.4bpp, when compared to the existing system architecture. The experimental findings are formulated in Table I.

TABLE I. COMPARISON OF ACCURACY PERCENTAGE OF THE CNN FOR WOW AND S-UNIWARD HAVING PAYLOADS OF 0.4 BPP AND 0.2 BPP

System	***Embedding Algorithm***	***Accuracy***	
		0.2 bpp	***0.4 bpp***
Proposed	WOW	89%	87.7%
	S-UNIWARD	88.9%	90.1%
Existing	WOW	80.3%	83.3%
	S-UNIWARD	73.6%	87.1%

SAMPLE OUTPUT

Process 1*:* An internet user opens a webpage of his choice, as depicted in Fig. 7. The page may or may not contain steganography in images. At this point, we check if there is any hidden data present in the images using our model which has been developed as an extension.

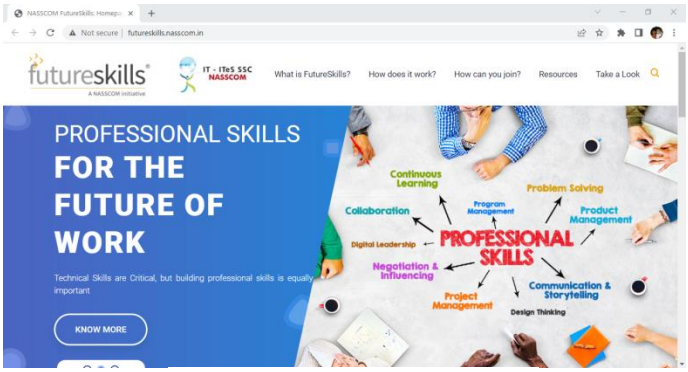

Fig. 7. Webpage opened by the user

Process 2: If the webpage contains any images that have any embedded code or malicious content, as exemplified in Fig. 8, the extension detects it with the help of CNN algorithm. As a result, the webpage will show an alert box displaying the message "Images displayed here may contain malicious content that could harm your system!".

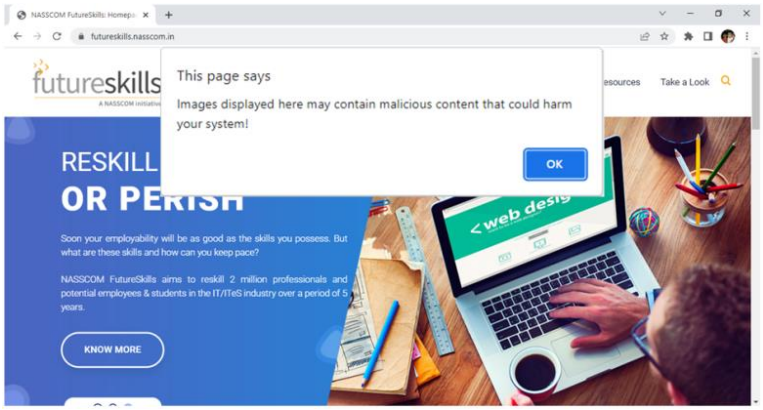

Fig. 8. Webpage displays alert

Process 3: If the webpage does not contain any images that have any embedded code or malicious content, the webpage will show an alert box displaying the message "Your browsing is secure", as illustrated in Fig. 9. With this message the user can be rest assured that his browsing is safe.

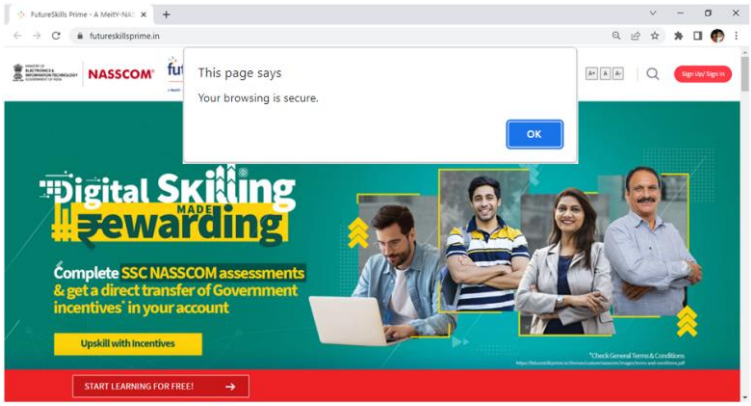

Fig. 9. Webpage that is secure

IX. CONCLUSION

Nowadays, safeguarding the transmission of personal data across network systems has become a difficult task due to cyber security concerns. There is a possibility that attackers will get access to the secret information, impart, divulge, or alter it, affecting its accessibility. In order to safeguard users' privacy from such threats, we have proposed a model based on Convolutional Neural Networks.

To identify the existence of concealed data in images, blind steganalysis can be used. Our proposed model, which is based on the Convolutional Neural Networks Algorithm (CNN) and Deep Learning, performs very well in discovering hidden data. The proposed model outperforms existing systems by achieving higher precision by testing across various payloads. We have tested for S-UNIWARD and WOW with the payload of both 0.2bpp and 0.4bpp. Our model has an accuracy of 89% and it surpasses the existing system which has an accuracy of 85%.

X. FUTURE SCOPE

The proposed system has been implemented via extensions provided in Google Chrome. As a future enhancement, the steganalyzer extension can be executed in other web browsers including Mozilla Firefox, Internet Explorer, Microsoft Edge, Safari, Brave, etc. Additionally, this system can also be included in mobile email applications such as Gmail, Yahoo Mail in order to scan the attached images that the user receives in Inbox. If there is any malicious data hidden in the mail, an alert can be generated in a similar way as the present system.

Some images may contain watermarking for copyright and security purposes, but this may hinder the steganalysis process and predict an erroneous output. As a potential future improvement, the proposed system should be able to differentiate the watermarking traces from steganographical features.

ACKNOWLEDGEMENT

We would like to convey our heartfelt gratitude to Mrs Anuratha K, as the guide, for her tremendous guidance and encouragement in carrying out this project work.

Authorized licensed use limited to: Jain University. Downloaded on 02/03/2023 at 09:45:04 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] You, Weike, Hong Zhang, and Xianfeng Zhao, "A Siamese CNN for image steganalysis," IEEE Transactions on Information Forensics and Security 16 (2020): 291-306.

[2] Boroumand, Mehdi, Mo Chen, and Jessica Fridrich, "Deep residual network for steganalysis of digital images," IEEE Transactions on Information Forensics and Security 14, no. 5 (2018): 1181-1193.

[3] Yang, Hongwei, Hui He, Weizhe Zhang, and Xiaochun Cao, "FedSteg: A federated transfer learning framework for secure image steganalysis," IEEE Transactions on Network Science and Engineering 8, no. 2 (2020): 1084-1094.

[4] Hermassi, Houcemeddine, "Blind image steganalysis using features extraction and machine learning," In 2021 18th International Multi-Conference on Systems, Signals & Devices (SSD), pp. 673-680. IEEE, 2021.

[5] Mustafa, M. Eslam, Mohamed A. Elshafey, and Mohamed M. Fouad, "Accuracy enhancement of a blind image steganalysis approach using dynamic learning rate-based CNN on GPUs," In 2019 10th IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS), vol. 1, pp. 28-33. IEEE, 2019.

[6] X. Liao, Q Wen, J Zhang, (2011), "A steganographic method for digital images with four-pixel differencing and modified LSB substitution," Journal of Visual Communication and Image Representation, 22(1), 1-8.

[7] Meenakumari and Rita Chhikara, "Blind image steganalysis using neural networks and wrapper feature selection," In 2017 International Conference on Computing, Communication and Automation (ICCCA), pp. 1065-1069. IEEE, 2017.

[8] Swain, (2014), "Digital image steganography using nine-pixel differencing and modified LSB substitution," Indian Journal of Science and Technology, 7(9), 1444-1450.

[9] J Yu, Xinpeng Zhang & Fengyong Li (2015), "Spatial steganalysis using redistributed residuals and diverse ensemble classifier," Multimedia Tools and Applications", 1-13.

[10] Harlap, Aaron, et al. "Pipedream: Fast and efficient pipeline parallel dnn training," arXiv preprint arXiv:1806.03377 (2018).

[11] L. Pibre, J. Pasquet, D. Ienco, and M. Chaumont, "Deep learning is a good steganalysis tool when embedding key is reused for different images, even if there is a cover source mismatch," Electronic Imaging, vol. 2016, no. 8, pp. 1–11, 2016.

[12] G. Xu, HZ. Wu and YQ. Shi, "Structural design of convolutional neural networks for steganalysis," IEEE Signal Processing Letters, vol. 23, no. 5, pp. 708–712, 2016.

[13] R Dhaya, "Light Weight CNN based Robust Image Watermarking Scheme for Security," Journal of Information Technology and Digital World 3, no. 2 (2021): 118-132.

[14] C. Fuji-Tsang and J. Fridrich. "Steganalyzing images of arbitrary size with CNNs," In A. Alattar and N. D. Memon, editors, Proceedings IS&T, Electronic Imaging, Media Watermarking, Security, and Forensics 2018, San Francisco, CA, January 29– February 1, 2018.

[15] J. Kodovsky, J. Fridrich and V. Holub, "Ensemble classifiers for steganalysis of digital media," IEEE Transactions on Information Forensics and Security, vol. 7, no. 2, pp. 432-444, 2012.

[16] Sorina Dumitrescu and Xiaolin Wu, "A new framework of lsb steganalysis of digital media," IEEE Transactions on Signal Processing, vol. 53, pp. 3936-3947, October 2005.

[17] Y. Qian, J. Dong, W. Wang and T. Tan, "Deep learning for steganalysis via convolutional neural networks" in Media Watermarking Security and Forensics 2015, International Society for Optics and Photonics, vol. 9409, pp. 94090, 2015.

[18] Shakya, Subarana, "An efficient security framework for data migration in a cloud computing environment," Journal of Artificial Intelligence 1, no. 01 (2019): 45-53.

Authorized licensed use limited to: Jain University. Downloaded on March 26,2026 at 09:45:04 UTC from IEEE Xplore. Restrictions apply.